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ROLL NO.: 22L-7651

SECTION: 4A (BS-EE)

ASSIGNMENT: 2

QUESTION 1:-

a) the armature currents are

$$\frac{76}{100} \times 33 = 25.3 \text{ A}$$

$$\frac{76}{100} \times 67 = 50.7 \text{ A}$$

$$\frac{76}{100} \times 100 = 76 \text{ A}$$

$$\frac{76}{100} \times 133 = 101.3 \text{ A}$$

If $I_A = 25.3 \text{ A}$
then

$$\begin{aligned} E_A &= V_t - I_A(R_A + R_s) \\ &= 240 - (25.3)(0.09 + 0.06) \\ &= 236.2 \text{ V} \end{aligned}$$

the magnetomotive force is $F = NI_A$

$$\begin{aligned} F &= (33)(25.3) \\ &= 835 \text{ A-turns} \end{aligned}$$

which produces a voltage E_{A_0} of 134 V at $n_0 = 900 \text{ r/min}$.

Now

$$\begin{aligned} n &= \frac{E_A}{E_{A_0}} \cdot n_0 \\ &= \frac{236.2}{134} (900) \end{aligned}$$

$$n = 1586 \text{ r/min}$$

the power converted from electrical to mechanical form is

$$P_{\text{conv}} = I_A E_A \Rightarrow (236.2)(25.3)$$

$$P_{\text{conv}} = 5976 \text{ W}$$

Since the rotational losses are ignored, this is also the output power of the motor. The induced torque is

$$\tau_{\text{ind}} = \frac{P_{\text{conv}}}{\omega_m}$$

$$= \frac{5976}{(1586) \left(\frac{2\pi \text{ rad}}{\text{r}} \right) \left(\frac{1}{60} \right)}$$

$$\tau_{\text{ind}} = 36 \text{ Nm}$$

If $I_A = 50.7 \text{ A}$, then
then

$$E_A = V_T - I_A (R_A + R_s)$$

$$= 240 - (50.7)(0.09 + 0.06)$$

$$E_A = 232.4 \text{ V}$$

Now

$$F = N I_A$$

$$= (33)(50.7)$$

$$= 1672 \text{ A.tums}$$

$$E_{A_0} = 252 \text{ V at } n_0 = 900 \text{ r/min}$$

the speed of the motor is

$$n = \frac{E_A}{E_{A_0}} n_0$$

$$= \frac{(232.4)(900)}{252}$$

$$197$$

$$n = 1062 \text{ r/min}$$

the power converted from electrical to mechanical form is

$$P_{\text{conv}} = E_A I_A \\ = (232.4)(50.7)$$

$$P_{\text{conv}} = 11780 \text{ W}$$

This is the output power of the motor as rotational losses are ignored

The induced torque is

$$T_{\text{ind}} = \frac{P_{\text{conv}}}{\omega_m} \\ = \frac{11780}{(1062)(2\pi)(\frac{1}{60})}$$

$$T_{\text{ind}} = 106 \text{ Nm}$$

If $I_A = 76 \text{ A}$,
then

$$E_A = V_t - I_A(R_A + R_s) \\ = 240 - (76)(0.09 + 0.06) \\ = 228.6 \text{ V}$$

Now

$$F = N \hat{I}_A \\ = (33)(76)$$

$$F = 2508 \text{ A.tums}$$

which produces a voltage E_{A0} of 229 V at $n_0 = 900 \text{ r/m}$.

The speed of motor is

$$n = \frac{E_A}{E_{A0}} n_0 \\ = \frac{228.6}{229}(900)$$

$$n = 899 \text{ r/min}$$

the power converted from electrical to mechanical form is

$$P_{conv} = E_A I_A \\ = (228.6)(76)$$

$$P_{conv} = 17,370 \text{ W}$$

Since the rotational losses are ignored, this is also the output power of the motor.

thus induced torque is

$$\tau_{ind} = \frac{P_{conv}}{\omega_m} \\ = \frac{17370}{(899)(2\pi)(1/60)}$$

$$\tau_{ind} = 185 \text{ Nm}$$

If $I_A = 101.3 \text{ A}$,
then

$$E_A = V_T - I_A(R_A + R_S) \\ = 240 - (101.3)(0.09 + 0.06)$$

$$E_A = 224.8 \text{ V}$$

Now

$$F = N I_A \\ = (33)(101.3)$$

$$F = 3343 \text{ A}\cdot\text{turns}$$

which produces a voltage E_{A0} of 252 V at $n_0 = 900 \text{ r/min}$.
the speed of motor is

$$n = \frac{E_A}{E_{A0}} \cdot n_0 \\ = \frac{(224.8)(900)}{(252)}$$

$$n = 803 \text{ r/min}$$

the power converted from electrical to mechanical form is

$$P_{\text{conv}} = I_A E_A$$

$$= (224.8)(101.3)$$

$$P_{\text{conv}} = 22770 \text{ W}$$

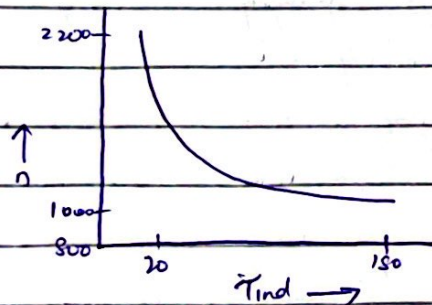
this is also the output power of motor.
the induced torque is

$$\tau_{\text{ind}} = \frac{P_{\text{conv}}}{\omega_m}$$

$$= \frac{22770}{(803)(2\pi)(1/60)}$$

$$\tau_{\text{ind}} = 271 \text{ Nm}$$

b) the terminal characteristics of the machine is



QUESTION 2:-

calculating the brush losses by substituting voltage drop 2V.

$$P_{\text{brush}} = (2)(500)$$

$$= 1000 \text{ W}$$

calculating the armature resistance

$$R_A = \frac{V_A}{I_A}$$

$$= \frac{16.3}{500}$$

$$R_A = 0.033 \Omega$$

Calculating the armature resistance

$$R_A = \frac{V_F}{I_F}$$

$$= \frac{440}{8.86}$$

$$R_A = 49.66 \Omega$$

At full load calculate the armature losses

$$P_A = I^2 R_A$$

$$= (500)^2 (0.033)$$

$$P_A = 8250 \text{ W}$$

Calculate the losses for field current

$$P_F = I^2 R$$

$$= (8.86)^2 (49.66)$$

$$P_F = 3898 \text{ W}$$

Calculate the copper losses for the motor

$$P_{cu} = I_A^2 R_A + V_F I_F$$

$$= (500)^2 (0.033) + (440)(8.86)$$

$$P_{cu} = 12.25 \text{ kW}$$

Now calculating the mechanical losses

$$P_{mech} = V_T I_A$$

$$= (440)(23.1)$$

$$P_{mech} = 10.164 \text{ kW}$$

calculating the input power

$$\begin{aligned} P_{in} &= (V_T)(I_L) \\ &= (440)(560) \\ &= 246.4 \text{ kW} \end{aligned}$$

calculating the stray loss power at full load

$$\begin{aligned} P_{\text{stray}} &= (P_{in}) \left(\frac{1}{100} \right) \\ &= (246.4 \text{ kW})(0.01) \\ &= 2.464 \text{ kW} \end{aligned}$$

calculating the output power

$$\begin{aligned} P_{out} &= P_{in} - P_{\text{brush}} - P_{cu} - P_{\text{core}} - P_{\text{mech}} - P_{\text{stray}} \\ P_{out} &= 218.53 \text{ kW} \end{aligned}$$

calculating the efficiency

$$\begin{aligned} \eta &= \frac{P_{out}}{P_{in}} \\ &= \frac{218.53 \text{ kW}}{246.4 \text{ kW}} (100\%) \end{aligned}$$

$$\boxed{\eta = 88.69\%}$$

thus, the DC motor's efficiency η is 88.69%.

QUESTION 3:-

a) If $R_{adj} = 120 \Omega$, then the field resistance is

$$\begin{aligned} R_F &= 200 + 120 \\ &= 320 \Omega \end{aligned}$$

$$\text{Field current} = I_F = \frac{V_T}{R_F + R_{adj}}$$

$$= \frac{240}{200 + 120}$$

$$= 0.75 \text{ A}$$

From the graph at 0.75 A, the voltage $E_{A0} = 260 \text{ V}$ at speed of 1200 r/min.

Actual $E_A = 240 \text{ V}$

the actual speed will be

$$n = \frac{E_A}{E_{A0}} n_0$$

$$= \frac{(240)(1200)}{260}$$

$$= 1107.692 \text{ r/min}$$

$$\boxed{n \approx 1107 \text{ r/min}}$$

b) At full load $I_A = I_L - I_F$

$$= 100 - 0.75$$

$$= 99.25 \text{ A}$$

$$E_A = V_T - I_A R_A$$

$$= 240 - (99.25)(0.14)$$

$$= 226.1 \text{ V}$$

the speed at full load is

$$n = \frac{226.1}{260} \times 1200$$

$$= 1043.53$$

$$\boxed{n \approx 1043 \text{ r/min}}$$

c) If R_{adj} is maximum at no load, R_T will be

$$R_T = 200 + 300 \\ = 500 \Omega$$

$$I_F = \frac{V_T}{R_F + R_{adj}} \Rightarrow \frac{240}{200 + 300} \\ I_F = 0.48 A$$

$$E_{A0} = 200 V \text{ at } n_0 = 1200 \text{ r/min}$$

Actual speed

$$n = \frac{E_A}{E_{A0}} n_0 \\ = \frac{240}{200} (1200) \\ = 1440 \text{ r/min}$$

If R_{adj} is minimum at no load, R_T will be

$$R_T = 0 + 200 \\ = 200 \Omega$$

$$I_F = \frac{V_T}{R_F + R_{adj}} \\ = \frac{240}{0 + 200}$$

$$I_F = 1.2 A$$

$$E_{A0} = 285 V \text{ at } n_0 = 1200 \text{ r/min}$$

$$n = \frac{E_A}{E_{A0}} \times n_m \\ = \frac{240}{285} \times 1200$$

$$= 1010.528$$

$$n = 1010 \text{ r/min}$$

QUESTION 5:-

1. At minimum speed

$$R_{adj-min} = 10 \Omega$$

$$I_{Fmin} = 0.25$$

$$E_{Amin} = 100V$$

$$\begin{aligned}\phi_{min} &= I_{Fmin} \times N_F \times (E_{Amin}/V) \\ &= 0.25 \times 1400 \times (100/250)\end{aligned}$$

$$\phi_{min} = 140 \text{ Weber}$$

$$\begin{aligned}N_{no-load-min} &= V / N_A \times \phi_{min} \\ &= 250 \\ &\quad (90 \times 140)\end{aligned}$$

$$N_{no-load-min} = 0.1786 \text{ rpm}$$

At maximum speed:

$$R_{adj-max} = 400 \Omega$$

$$I_{Fmax} = 1.25A$$

$$E_{Amax} = 290V$$

$$\begin{aligned}\phi_{max} &= I_{Fmax} \times N_F \times (E_{Amax}/V) \\ &= 1.25 \times 1400 \times (290/250) \\ &= 2030 \text{ Wb}\end{aligned}$$

$$\begin{aligned}N_{no-load-max} &= V / N_A \times \phi_{max} \\ &= 250 \\ &\quad 90 \times 2030\end{aligned}$$

$$N_{no-load-max} = 0.0159 \text{ rpm}$$

$\therefore$ the range is 0.0159 to 0.1786 rpm.

2. Full load speed when $R_{adj} = 100 \Omega$

$$R_{adj} = 100 \Omega$$

$$I_F = 0.75$$

$$E_A = V$$

$$\begin{aligned}\phi &= I_F \times N_F \times (E_A / V) \\ &= 0.75 \times 1400 \times \left(\frac{250}{250} \right)\end{aligned}$$

$$\phi = 1050 \text{ Wb}$$

$$\begin{aligned}N_{100} &= \left(\frac{V - I_A \times R_A}{N_A \times \phi} \right) \\ &= \frac{250 - 120 \times 0.18}{90 \times 1050}\end{aligned}$$

$$N_{100} = 0.0119 \text{ rpm}$$

Full load speed when $R_{adj} = 200 \Omega$

$$R_{adj} = 200 \Omega$$

$$I_F = 1 \text{ A}$$

$$E_A = V$$

$$\begin{aligned}\phi &= I_F \times N_F \times (E_A / V) \\ &= 1 \times 1400 \times \left(\frac{250}{250} \right)\end{aligned}$$

$$\phi = 1400 \text{ Wb}$$

$$\begin{aligned}N_{200} &= \left(\frac{V - I_A \times R_A}{N_A \times \phi} \right) \\ &= \frac{250 - 120 \times (0.18)}{90 \times 1400}\end{aligned}$$

$$N_{200} = 0.0022 \text{ rpm}$$

Full load speed when $R_{adj} = 300 \Omega$

$$R_{adj} = 300 \Omega$$

$$I_F = 1.25 \text{ A}$$

$$E_A = V$$

$$\phi = I_F \times N_F \times (E_A / V)$$

$$= 1.25 \times 1400 \times (250 / 250)$$

$$\phi = 1750 \text{ Wb}$$

$$N_{300} = \frac{(V - I_A)(R_A)}{N_A \times \phi}$$

$$= \frac{250 - (120 \times 0.18)}{90 \times 1750}$$

$$N_{300} = 0.0012 \text{ rpm}$$

3. Induced torque when $R_{adj} = 100 \Omega$

$$\phi = 1050 \text{ Wb}$$

$$T_{100} = \frac{\phi \times I_A}{2\pi}$$

$$= \frac{1050 \times 120}{2\pi}$$

$$T_{100} = 1995.98 \text{ Nm}$$

Induced torque when $R_{adj} = 200 \Omega$

$$\phi = 1400 \text{ Wb}$$

$$T_{200} = \frac{\phi \times I_A}{2\pi}$$

$$= \frac{1400 \times 120}{2\pi}$$

$$T_{200} = 2679.61 \text{ Nm}$$

Induced Torque when $R_{adj} = 300 \Omega$

$$\phi = 1750 \text{ Wb}$$

$$T_{300} = \frac{\phi \times I_A}{2\pi}$$

$$\tau_{300} = \frac{1750 \times 120}{2\pi}$$

$$\tau_{300} = 3319.46 \text{ Nm}$$

4 Output power when $R_{adj} = 100 \Omega$
 $N = 0.0119 \text{ rpm}$

$$P_{100} = \frac{\tau_{100} \times N}{60 \times 1000}$$

$$= \frac{1995.98 \times 0.0119}{60 \times 1000}$$

$$P_{100} = 0.00318 \text{ kW}$$

Load torque when $R_{adj} = 100 \Omega$

$$\tau_{L100} = \frac{(\text{core losses} + \text{Mechanical losses})}{N}$$

$$= \frac{500 + 400}{0.0119}$$

$$\tau_{L100} = 84033.61 \text{ Nm}$$

Output power when $R_{adj} = 200 \Omega$
 $N = 0.0022 \text{ rpm}$

$$P_{200} = \frac{\tau_{200} \times N}{60 \times 1000}$$

$$= \frac{2679.61 \times 0.0022}{60 \times 1000}$$

$$P_{200} = 0.00127 \text{ kW}$$

Load torque when $R_{adj} = 200 \Omega$

$$\tau_{L200} = \frac{(500 + 400)}{0.0022}$$

~~Answer~~

$$\tau_{1200} = 409090.91 \text{ Nm}$$

Output power when $R_{adj} = 300 \Omega$

$$N = 0.0012 \text{ rpm}$$

$$P_{300} = \tau_{300} \times N$$

$$= \frac{60 \times 1000}{60 \times 1000} (3319.46) (0.0012)$$

$$P_{300} = 0.00199 \text{ kW}$$

Load torque when $R_{adj} = 300 \Omega$

$$\tau_{2300} = \frac{500 + 400}{0.0012}$$

$$\tau_{2300} = 750000 \text{ Nm}$$

QUESTION 4:-

a) To find the internally generated voltage at a full current of 1A, we can use the equation

$$E_A = (K \times I_F) - (R_{Fida} \times I_F)$$

Given that the rated terminal voltage is 240V, we can calculate the value of K as follows

$$K = E_A / I_F$$

$$= 240 / 1$$

$$K = 240 \text{ V/A}$$

Now

$$E_A = (240 \times 1) - (0.45 \times 1)$$

$$= 240 - 0.45$$

$$E_A = 239.55 \text{ V}$$

therefore, the internally generated voltage as a field current of 1 A is approx. 239.55V.

b) the value of adjustable resistance can be determined by subtracting the field resistance from total resistance.

$$R_t = R_f + R_{adj}$$

$$R_f = 100 \Omega, \text{ so}$$

$$R_{adj} = R_t - R_f \\ = 700 - 100$$

$$R_{adj} = 600 \Omega$$

therefore, the value of adjustable resistance connected in series with the field coil is 600Ω .

c) To find the terminal voltage when load current is 45 A, we can use the equation

$$V_t = V_g - (I_L \times R_a)$$

so

$$V_t = 240 - (45 \times 0.45)$$

$$V_t = 219.75 \text{ V}$$

d) the output power can be calculated as

$$\text{Output power} = V_t \times I_L$$

$$= 219.75 \times 45 \text{ A}$$

$$\text{Output power} = 9888.75 \text{ W}$$

therefore input power of the generator is

$$\text{Input power} = \text{Output power} + \text{core losses}$$

$$= 9888.75 + 400$$

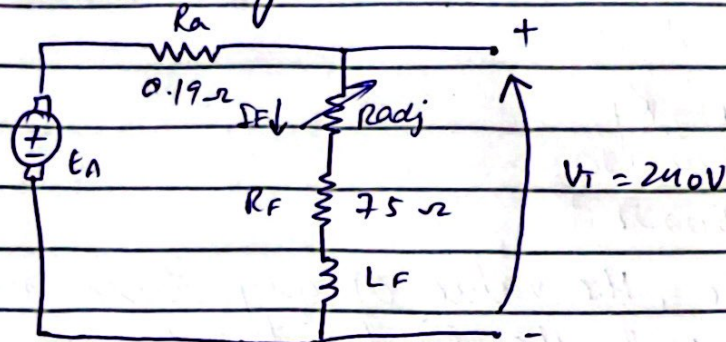
$$\text{Input power} = 10288.75 \text{ W}$$

$$\text{or } 10.29 \text{ kW}$$

TEXT BOOK PROBLEMS

PROBLEM 1:-

the circuit diagram is



At no load,

$$E_A = V_T = 240 \text{ V}$$

When R_{adj} is adjusted to 175Ω , the field current of the shunt motor changes.

the field current is

$$I_F = \frac{V_T}{R_{adj} + R_F}$$

$$= \frac{240}{175 + 75}$$

$$= \frac{240}{250}$$

$$I_F = 0.96$$

therefore corresponding to $I_F = 0.96$, $E_A = 241 \text{ V}$.

So

the no-load speed of motor at 240V using the below relation

$$\frac{E_{A \text{ no-load}}}{E_A} = \frac{n_{\text{no-load}}}{n}$$

$$\frac{240}{241} = \frac{n_{\text{no-load}}}{1800}$$

$$n_{\text{no-load}} = \left(\frac{240}{241} \right) \times 1800$$

$$n_{\text{no-load}} = 1792.53 \text{ r/min}$$

PROBLEM 2:-

Using the circuit in problem 1,
finding full load armature current I_A using KCL

$$I_A = I_{L \text{ rated}} - I_F \rightarrow \textcircled{1}$$

Finding field current I_F

$$I_F = \frac{V_T}{R_{adj} + R_F}$$

$$I_F = \frac{240}{175 + 75}$$
$$= \frac{240}{250}$$

$$I_F = 0.96 \text{ A}$$

Now solving eq ①

$$I_A = 110 - 0.96$$
$$= 109.04 \text{ A}$$

Now finding the full load internal generated

voltage E_{Afl} using KVL,

$$E_{Afl} = V_T - I_A R_A$$

$$E_{Afl} = 240 - (109.04)(0.19)$$

$$E_{Afl} = 219.29 \text{ V}$$

M

$$E_A \propto n$$

Now finding the full load speed of motor

$$\frac{E_A}{E_{Afl}} = \frac{n}{n_{fl}}$$

$$\frac{240}{219.29} = \frac{n_{fl}}{1800}$$

$$n_{fl} = \left(\frac{219.29}{240} \right) \times 1800$$

$$n_{fl} = 1637.85 \text{ r/min}$$

the expression of speed regulation of motor is

$$SR = \left(\frac{n_{no-load} - n_{fl}}{n_{fl}} \right) \times 100 \rightarrow (3)$$

here

$$E_A = V_T = 240 \text{ V}$$

Now finding the no load speed of motor at 240 V is

$$\frac{E_{A \text{ no-load}}}{E_A} = \frac{n_{no-load}}{n}$$

$$\frac{240}{241} = \frac{n_{no-load}}{1800}$$

$$n_{no-load} = \left(\frac{240}{241} \right) \times 1800$$

$$n_{no-load} = 1792.53 \text{ r/min}$$

$$n_{no-load} = 1792.53 \text{ r/min}$$

substituting values in (3)

$$SR = \left(\frac{1792.53 - 1637.85}{1637.85} \right) \times 100$$
$$= \left(\frac{154.68}{1637.85} \right) \times 100$$

$$SR = 9.44\%$$

PROBLEM 3:-

Using the circuit in problem 1, when R_{adj} is adjusted to 250Ω , the field current changes. Finding the field current during full load,

$$I_F = \frac{V_T}{R_{adj} + R_F}$$
$$= \frac{240}{175 + 75}$$
$$= \frac{240}{250}$$

$$I_F = 0.96A$$

Now finding full load armature current using KCL

$$I_A = I_{load} - I_F$$
$$= 110 - 0.96$$

$$I_A = 109.04A$$

From the circuit diagram, finding the $E_{A\mu}$ using KVL

$$E_{A\mu} = V_T - I_A R_A$$

$$E_{A\mu} = 240 - (109.04)(0.19)$$

$$E_{A\mu} = 219.29V$$

therefore, the full load speed of motor is

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0}$$

$$\frac{219.29}{240} = \frac{n}{1800}$$

$$n_m = \left(\frac{219.29}{240} \right) \times 1800$$

$$n_m = 1644.67 \text{ r/min}$$

$$n_m = 1644.67 \text{ r/min}$$

Now with $R_{adj} = 250$

$$I_F = \frac{240}{250 + 75}$$

$$= \frac{240}{325}$$

$$I_F = 0.738 \text{ A}$$

Now finding n_m

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0}$$

$$\frac{219.29}{210} = \frac{n}{1800}$$

$$n_m = \left(\frac{219.29}{210} \right) \times 1800$$

$$n_m = 1879.71 \text{ r/min}$$

$$n_m = 1879.71 \text{ r/min}$$

Hence the resistance R_{adj} increases the speed of motor n also increases

PROBLEM 5

Finding the field current when $R_{adj} = 100 \Omega$

$$I_F = \frac{V_T}{R_{adj} + R_F}$$
$$= \frac{240}{100 + 75}$$

$$I_F = 1.37 A$$

As

$$E_A \propto n$$

So

$$\frac{E_A}{E_{A_0}} = \frac{n_{min}}{n_0}$$

$$\frac{240}{272} = \frac{n_{min}}{1800}$$

$$n_{min} = \left(\frac{240}{272} \right) \times 1800$$

$$n_{min} = 1588.23 \text{ r/min}$$

Now finding the field current when $R_{adj} = 400 \Omega$

$$I_F = \frac{V_T}{R_{adj} + R_F}$$
$$= \frac{240}{400 + 75}$$

$$= \frac{240}{475}$$

$$I_F = 0.505 A$$

Now finding no load speed.

$$\frac{E_A}{E_{A0}} = \frac{n_{max}}{n_0}$$

$$\frac{240}{167} = \frac{n_{max}}{1800}$$

$$n_{max} = \left(\frac{240}{167} \right) \times 1800$$

$$n_{max} = 2586.82 \text{ r/min}$$

PROBLEM 6:-

When the motor is started by connecting directly to the power supply, the expression of starting current of the motor is given below

$$I_{L \text{ start}} = \frac{V_T}{R_A}$$

$$= \frac{240}{0.19}$$

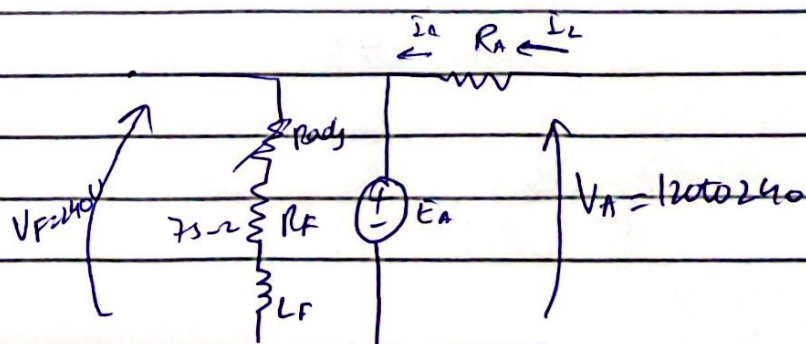
$$I_{L \text{ start}} = 1263.15 \text{ A}$$

the full load of the motor is given as 110 and which is

$$\frac{1263.15}{110} = 11.48 \text{ times higher than the full}$$

load current.

PROBLEM 8:-



a) When armature voltage of $V_A = 120V$, finding the no-load speed of the motor.

At no-load,

$$E_A = V_A$$

Firstly, calculating the I_F

$$I_F = \frac{V_F}{R_{adj} + R_F}$$

$$I_F = \frac{240}{175 + 75}$$

$$= \frac{240}{250}$$

$$I_F = 0.96A$$

Now finding minimum no-load speed of motor at $V_A = 120V$.

$$\frac{V_A}{E_{A_0}} = \frac{n_m}{n_0}$$

$$\frac{120}{240} = \frac{n_m}{1800}$$

$$n_m = \left(\frac{120}{240} \right) \times 1800$$

$$n_m = 900 \text{ r/min}$$

b) Now finding the no-load speed of the motor at $E_A = V_A = 180V$

$$\frac{V_A}{E_{A_0}} = \frac{n_m}{n_0}$$

$$\frac{180}{240} = \frac{n_m}{1800}$$

$$n_m = \left(\frac{120}{240} \right) \times 1800$$

$$n_m = 1350 \text{ r/min}$$

c) Finding the no load speed of motor at $E_A = V_A = 240 \text{ V}$

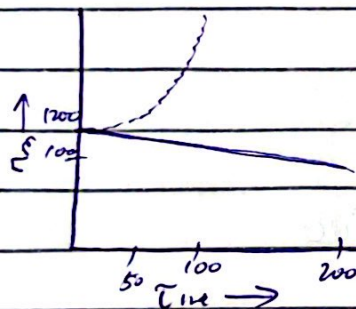
$$\frac{V_A}{E_{A_0}} = \frac{n_m}{n_0}$$

$$\frac{240}{240} = \frac{n_m}{1800}$$

$$n_m = 1800 \text{ r/min}$$

PROBLEM 7:-

the plot is given below



PROBLEM 9:-

a) the maximum speed of the motor is obtained by reducing the field current to a low value and by changing the adjustable resistance to max value and supply max voltage across the armature, we can achieve the reducing field current and maximize the speed.

the max value of adjustable resistance is

$R_{adj} = 400\Omega$, so the I_F is

$$I_F = \frac{V_F}{R_{adj} + R_F}$$

$$= \frac{240}{400 + 75}$$

$$I_F = 0.50526 A$$

From magnetization current I_F , the internal generated voltage $E_{A0} = 167 V$, then

$$n_0 = 1800 \text{ r/min}$$

Here

$$V_T = 240 V$$

speed required by motor is

$$n_m = \left(\frac{E_A}{E_{A0}} \right) n_0$$

$$= \frac{240}{167} \times 1800$$

$$n_m = 2586.826 \text{ r/min}$$

b) the minimum speed of the motor is obtained by increasing the field current to a high value. By changing the R_{adj} to minimum value and apply minimum voltage across armature, we can achieve min speed

the min. value of R_{adj} is 100Ω .

the field current is

$$I_F = \frac{V_F}{R_{adj} + R_F}$$

$$= \frac{240}{100 + 75}$$

$$I_F = 1.3714 \text{ A}$$

the internal generated voltage at no load is

$$E_{A0} = 271 \text{ V}, \quad n_0 = 1800 \text{ r/min}$$

$$V_T = E_A = 1200$$

the min. no load speed is

$$n_m = \left(\frac{E_A}{E_{A0}} \right) n_0$$

$$= \left(\frac{1200}{271} \right) (1800)$$

$$n_m = 797.04 \text{ r/min}$$

c) the power input to the motor is

$$P_{in} = V_T I_{L, \text{rated}}$$

$$P_{in} = (240)(110)$$

$$= 26400$$

the rated field current is

$$I_F = \frac{V_F}{R_F}$$

$$= \frac{240}{75}$$

$$= 3.2 \text{ A}$$

the losses on the shunt field are

$$P_F = I_F^2 R_F$$

$$= (3.2)^2 (7.5)$$

$$P_F = 768 \text{ W}$$

Copper losses $P_A = I_A^2 R_A$

$R_A = 0.19 \Omega$

$R_A = I_{Lrated} - I_F$

$= 110 - 3.2$

$= 106.8 A$

$P_A = (106.8)^2 (0.19)$

$P_A = 2167.185 W$

the drop in brush voltage is 2V

$V_{BD} = 2 V$

$P_{brush} = V_{BD} \times I_A$

$= 2 \times 106.8$

$= 213.6 W$

rotational losses = core losses = $P_{rot} = P_{cor} = 3550 W$

$P_{stray} = (0.01) P_{in}$

$= (0.01)(26400)$

$= 264 W$

Output power of motor is

$P_{out} = P_{in} - P_A - P_{brush} - P_{core} - P_F - P_{stray}$

$= 26400 - 2167.1856 - 213.6 - 3550 - 768 - 264$

264

$P_{out} = 19437.2144 W$

$E \text{ efficiency} = \frac{19437.2144}{26400} \times 100\%$

$\text{efficiency} = 73.62\%$

PROBLEM 10:-

a) Given

$$R_{adj} = 175 \Omega$$

At no-load conditions, $E_A = V_T = 240 \text{ V}$

the field current is

$$I_F = \frac{V_T}{R_{adj} + R_F}$$

$$\begin{aligned} &= \frac{240}{175 + 75} \\ &= \frac{240}{250} \end{aligned}$$

$$I_F = 0.960 \text{ A}$$

The field current would produce an internal generated voltage E_{Ao} of 241 at speed of 1200 r/min. Therefore the speed n with a voltage of 240 would be

$$\begin{aligned} n &= \left(\frac{E_A}{E_{Ao}} \right) n_o \\ &= \left(\frac{240}{241} \right) (1200) \end{aligned}$$

$$\boxed{n = 1195 \text{ r/min}}$$

b) At full load conditions, the armature current is

$$\begin{aligned} I_A &= I_L - I_F \\ &= I_L - \frac{V_T}{R_{adj} + R_F} \\ &= 110 - 0.96 \\ &= 109 \text{ A} \end{aligned}$$

the internal generated voltage E_A is

$$\begin{aligned} E_A &= V_T - I_A (R_A + R_S) \\ &= 240 - 109(0.21) \\ &= \cancel{217.11} \\ &= 217.11 \text{ V} \end{aligned}$$

therefore

$$\begin{aligned} n &= \left(\frac{E_A}{E_{A0}} \right) n_0 \\ &= \left(\frac{217.11}{279} \right) (1200) \end{aligned}$$

$$\boxed{n = 934 \text{ r/min}}$$

c) the speed regulation is

$$SR = \frac{1195 - 934}{934} \times 100 = \left(\frac{n_{n1} - n_{n2}}{n_{n2}} \right) \times 100$$

$$\boxed{SR = 29.9\%}$$

d) the plot is given below

